

(e)

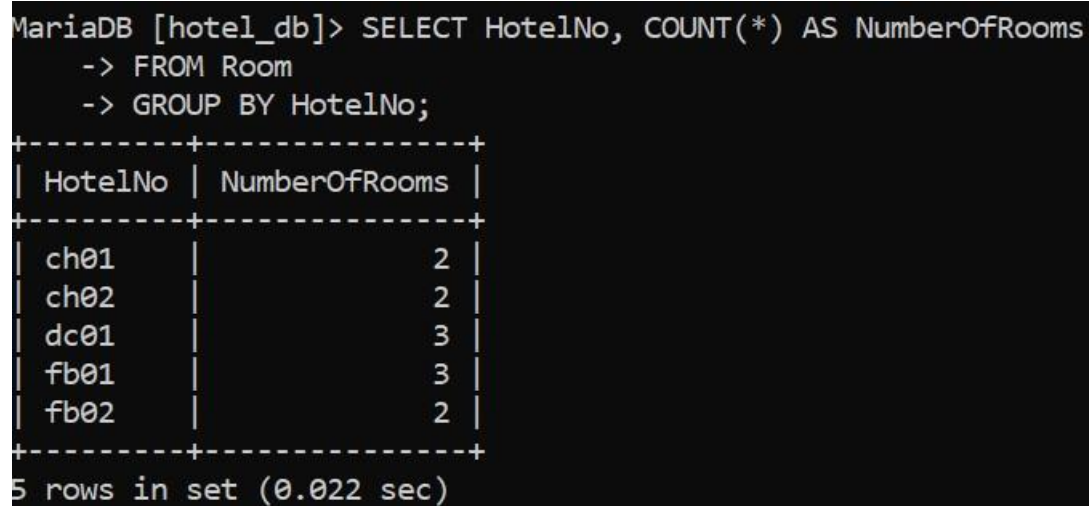
i. List the number of rooms in each hotel.

Query:

```
SELECT HotelNo, COUNT(*) AS NumberOfRooms
```

```
FROM Room
```

```
GROUP BY HotelNo;
```



The screenshot shows a MariaDB terminal window with the following content:

```
MariaDB [hotel_db]> SELECT HotelNo, COUNT(*) AS NumberOfRooms
-> FROM Room
-> GROUP BY HotelNo;
```

| HotelNo | NumberOfRooms |
|---------|---------------|
| ch01    | 2             |
| ch02    | 2             |
| dc01    | 3             |
| fb01    | 3             |
| fb02    | 2             |

5 rows in set (0.022 sec)

ii. List the number of rooms in each hotel in London.

Query:

```
SELECT Hotel.HotelNo, COUNT(Room.RoomNo) AS NumberOfRooms
```

```
FROM Room
```

```
JOIN Hotel ON Room.HotelNo = Hotel.HotelNo
```

```
WHERE Hotel.City = 'London'
```

```
GROUP BY Hotel.HotelNo;
```

```
MariaDB [hotel_db]> SELECT Hotel.HotelNo, COUNT(Room.RoomNo) AS NumberOfRooms
-> FROM Room
-> JOIN Hotel ON Room.HotelNo = Hotel.HotelNo
-> WHERE Hotel.City = 'London'
-> GROUP BY Hotel.HotelNo;
```

```
+-----+-----+
| HotelNo | NumberOfRooms |
+-----+-----+
| ch01    | 2             |
| ch02    | 2             |
| fb01    | 3             |
+-----+-----+
3 rows in set (0.001 sec)
```

iii. What is the average number of bookings for each hotel in April?

Query:

```
SELECT HotelNo, AVG(BookingsCount) AS AverageBookings
```

```
FROM (
```

```
    SELECT HotelNo, COUNT(*) AS BookingsCount
```

```
    FROM Booking
```

```
    WHERE MONTH(DateFrom) = 4
```

```
    GROUP BY HotelNo, GuestNo
```

```
) AS MonthlyBookings
```

```
GROUP BY HotelNo;
```

```
MariaDB [hotel_db]> SELECT HotelNo, AVG(BookingsCount) AS AverageBookings
-> FROM (
->     SELECT HotelNo, COUNT(*) AS BookingsCount
->     FROM Booking
->     WHERE MONTH(DateFrom) = 4
->     GROUP BY HotelNo, GuestNo
-> ) AS MonthlyBookings
-> GROUP BY HotelNo;
```

```
+-----+-----+
| HotelNo | AverageBookings |
+-----+-----+
| ch01    | 1.0000          |
| ch02    | 1.0000          |
| fb01    | 1.0000          |
| fb02    | 1.0000          |
+-----+-----+
4 rows in set (0.002 sec)
```

iv. What is the most commonly booked room type for each hotel in London?

Query :

```
SELECT h.HotelNo, r.Type, COUNT(*) AS BookingCount
FROM Booking b
JOIN Room r ON b.RoomNo = r.RoomNo AND b.HotelNo = r.HotelNo
JOIN Hotel h ON b.HotelNo = h.HotelNo
WHERE h.City = 'London'
GROUP BY h.HotelNo, r.Type
ORDER BY h.HotelNo, BookingCount DESC;
```

```
MariaDB [hotel_db]> SELECT h.HotelNo, r.Type, COUNT(*) AS BookingCount
-> FROM Booking b
-> JOIN Room r ON b.RoomNo = r.RoomNo AND b.HotelNo = r.HotelNo
-> JOIN Hotel h ON b.HotelNo = h.HotelNo
-> WHERE h.City = 'London'
-> GROUP BY h.HotelNo, r.Type
-> ORDER BY h.HotelNo, BookingCount DESC;
```

| HotelNo | Type   | BookingCount |
|---------|--------|--------------|
| ch01    | family | 1            |
| ch02    | double | 1            |
| fb01    | double | 2            |
| fb01    | single | 2            |
| fb01    | family | 1            |

5 rows in set (0.002 sec)

v. What is the lost income from unoccupied rooms at each hotel today?

Query:

```
SELECT h.HotelNo, h.HotelName, SUM(r.Price) AS LostIncome
FROM Hotel h
JOIN Room r ON h.HotelNo = r.HotelNo
LEFT JOIN (
    SELECT b.HotelNo, b.RoomNo
    FROM Booking b
    WHERE CURDATE() BETWEEN b.DateFrom AND IFNULL(b.DateTo, CURDATE())
```

) occupied\_rooms ON r.HotelNo = occupied\_rooms.HotelNo AND r.RoomNo = occupied\_rooms.RoomNo

WHERE occupied\_rooms.RoomNo IS NULL

GROUP BY h.HotelNo, h.HotelName;

```
MariaDB [hotel_db]> SELECT h.HotelNo, h.HotelName, SUM(r.Price) AS LostIncome
-> FROM Hotel h
-> JOIN Room r ON h.HotelNo = r.HotelNo
-> LEFT JOIN (
->     SELECT b.HotelNo, b.RoomNo
->     FROM Booking b
->     WHERE CURDATE() BETWEEN b.DateFrom AND IFNULL(b.DateTo, CURDATE())
-> ) occupied_rooms ON r.HotelNo = occupied_rooms.HotelNo AND r.RoomNo = occupied_rooms.RoomNo
-> WHERE occupied_rooms.RoomNo IS NULL
-> GROUP BY h.HotelNo, h.HotelName;
```

| HotelNo | HotelName     | LostIncome |
|---------|---------------|------------|
| ch01    | Omni Shoreham | 29.99      |
| ch02    | Phoenix Park  | 25.00      |
| dc01    | Latham        | 53.00      |
| fb01    | Grosvenor     | 48.00      |
| fb02    | Watengate     | 58.00      |

5 rows in set (0.025 sec)

(f) Backup the database hotel\_db as bkphotel\_db.sql in your desktop and drop the hotel\_db database and restore it with the newhotel\_dbname.

Query:

DROP DATABASE hotel\_db;

CREATE DATABASE newhotel\_db;